Primetrade.ai

Python Developer (Trading Systems) internship

BINANCE FUTURES TRADING BOT
Technical Report

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Application Task: Junior Python Developer

PROJECT OVERVIEW

Developed a CLI-based trading bot for Binance USDT-M Futures Testnet that executes market orders, limit orders, and TWAP algorithmic trading.

Development Time: ~30 minutes

GitHub: https://github.com/Vedant-D04/Binance-Trading-Bot.git

Tech Stack:
- Python 3.11
- python-binance
- Typer (CLI)
- Rich (formatting)

PROBLEM STATEMENT

Build a CLI application that can:
1. Place market orders (BUY/SELL) on Binance Futures Testnet
2. Place limit orders with price specification
3. Validate all inputs before API submission
4. Log all operations with timestamps
5. Handle errors gracefully

Bonus: Implement advanced order types

APPROACH

1. Modular Architecture
   - client.py: Binance API wrapper
   - orders.py: Order logic

- validators.py: Input validation
- cli.py: User interface

2. Validation Strategy
- Fetch exchange filters (LOT_SIZE, PRICE_FILTER)
- Validate before API calls
- Dynamic precision rounding

3. Error Handling
- Try-catch on all API calls
- Clear error messages
- Comprehensive logging

4. Bonus Feature: TWAP
- Split large orders into chunks
- Execute over time intervals
- Minimize market impact

TECHNICAL IMPLEMENTATION

1. MARKET ORDERS
Command: python cli.py market BTCUSDT BUY 0.001

Flow:
- Validate symbol exists
- Validate quantity meets minimum
- Execute at current market price
- Log response (orderId, status, avgPrice)

2. LIMIT ORDERS
Command: python cli.py limit ETHUSDT SELL 0.01 3500

Flow:
- All market order validations
- Validate price precision
- Place order on order book
- Return order ID for tracking

3. TWAP EXECUTION (Bonus)
Command: python cli.py twap BTCUSDT BUY 0.05 15 5

Logic:

- Total: 0.05 BTC over 15 minutes in 5 intervals
- Executes: 0.01 BTC every 3 minutes
- Calculates average execution price
- Logs each interval

4. VALIDATION SYSTEM
- Real-time exchange filter checking
- LOT_SIZE validation (min/max/step)
- PRICE_FILTER validation (precision)
- Prevents rejected orders

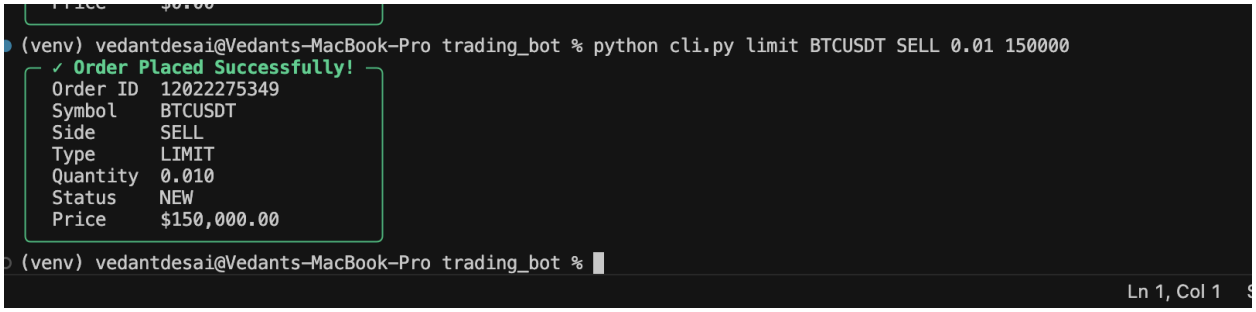
5. ERROR HANDLING
Handled Cases:
- Invalid symbol
- Insufficient balance
- Invalid quantity/price
- Network failures
- API rate limits

OUTPUT SCREENSHOTS

[Screenshot 1: Successful Market Order]
$ python cli.py market BTCUSDT BUY 0.01

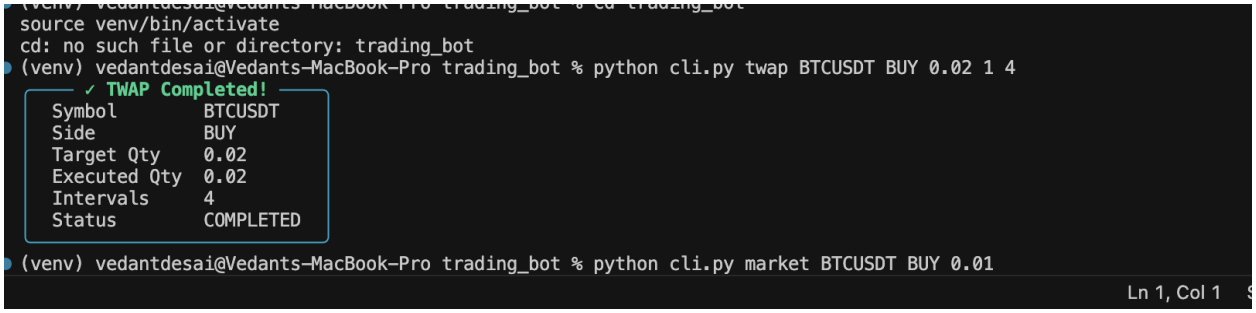


[Screenshot 2: Successful Limit Order]
$ python cli.py limit BTCUSDT SELL 0.01 150000

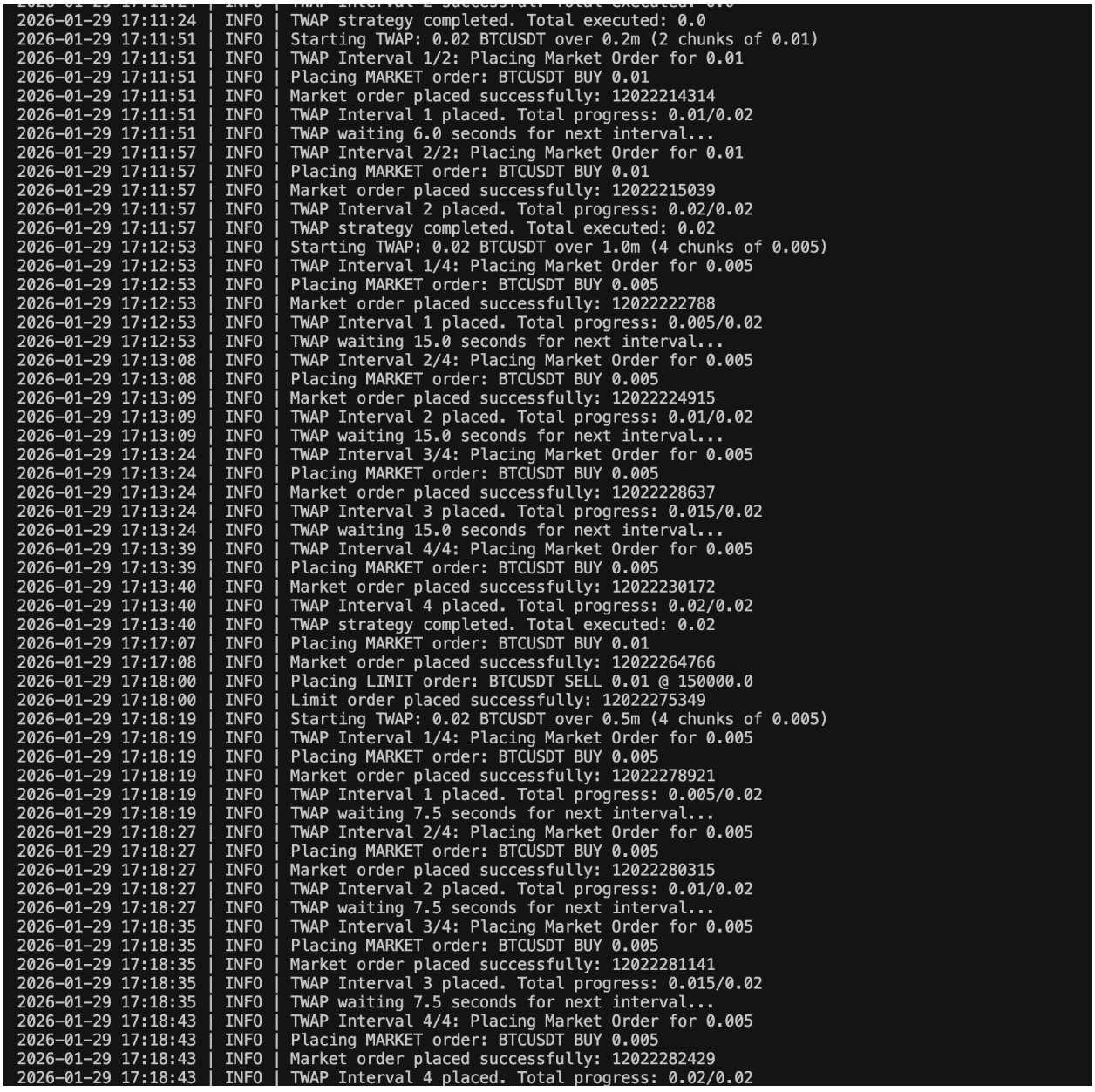


[Screenshot 3: TWAP Execution]
$ python cli.py twap BTCUSDT BUY 0.02 1 4

TWAP Execution Started



[Screenshot 4: bot.log excerpt]



```
2026-01-29 17:11:24 | INFO | TWAP strategy completed. Total executed: 0.0
2026-01-29 17:11:51 | INFO | Starting TWAP: 0.02 BTCUSDT over 0.2m (2 chunks of 0.01)
2026-01-29 17:11:51 | INFO | TWAP Interval 1/2: Placing Market Order for 0.01
2026-01-29 17:11:51 | INFO | Placing MARKET order: BTCUSDT BUY 0.01
2026-01-29 17:11:51 | INFO | Market order placed successfully: 12022214314
2026-01-29 17:11:51 | INFO | TWAP Interval 1 placed. Total progress: 0.01/0.02
2026-01-29 17:11:51 | INFO | TWAP waiting 6.0 seconds for next interval...
2026-01-29 17:11:57 | INFO | TWAP Interval 2/2: Placing Market Order for 0.01
2026-01-29 17:11:57 | INFO | Placing MARKET order: BTCUSDT BUY 0.01
2026-01-29 17:11:57 | INFO | Market order placed successfully: 12022215039
2026-01-29 17:11:57 | INFO | TWAP Interval 2 placed. Total progress: 0.02/0.02
2026-01-29 17:11:57 | INFO | TWAP strategy completed. Total executed: 0.02
2026-01-29 17:12:53 | INFO | Starting TWAP: 0.02 BTCUSDT over 1.0m (4 chunks of 0.005)
2026-01-29 17:12:53 | INFO | TWAP Interval 1/4: Placing Market Order for 0.005
2026-01-29 17:12:53 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:12:53 | INFO | Market order placed successfully: 12022222788
2026-01-29 17:12:53 | INFO | TWAP Interval 1 placed. Total progress: 0.005/0.02
2026-01-29 17:12:53 | INFO | TWAP waiting 15.0 seconds for next interval...
2026-01-29 17:13:08 | INFO | TWAP Interval 2/4: Placing Market Order for 0.005
2026-01-29 17:13:08 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:13:09 | INFO | Market order placed successfully: 12022224915
2026-01-29 17:13:09 | INFO | TWAP Interval 2 placed. Total progress: 0.01/0.02
2026-01-29 17:13:09 | INFO | TWAP waiting 15.0 seconds for next interval...
2026-01-29 17:13:24 | INFO | TWAP Interval 3/4: Placing Market Order for 0.005
2026-01-29 17:13:24 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:13:24 | INFO | Market order placed successfully: 12022228637
2026-01-29 17:13:24 | INFO | TWAP Interval 3 placed. Total progress: 0.015/0.02
2026-01-29 17:13:24 | INFO | TWAP waiting 15.0 seconds for next interval...
2026-01-29 17:13:39 | INFO | TWAP Interval 4/4: Placing Market Order for 0.005
2026-01-29 17:13:39 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:13:40 | INFO | Market order placed successfully: 12022230172
2026-01-29 17:13:40 | INFO | TWAP Interval 4 placed. Total progress: 0.02/0.02
2026-01-29 17:13:40 | INFO | TWAP strategy completed. Total executed: 0.02
2026-01-29 17:17:07 | INFO | Placing MARKET order: BTCUSDT BUY 0.01
2026-01-29 17:17:08 | INFO | Market order placed successfully: 12022264766
2026-01-29 17:18:00 | INFO | Placing LIMIT order: BTCUSDT SELL 0.01 @ 150000.0
2026-01-29 17:18:00 | INFO | Limit order placed successfully: 12022275349
2026-01-29 17:18:19 | INFO | Starting TWAP: 0.02 BTCUSDT over 0.5m (4 chunks of 0.005)
2026-01-29 17:18:19 | INFO | TWAP Interval 1/4: Placing Market Order for 0.005
2026-01-29 17:18:19 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:18:19 | INFO | Market order placed successfully: 12022278921
2026-01-29 17:18:19 | INFO | TWAP Interval 1 placed. Total progress: 0.005/0.02
2026-01-29 17:18:19 | INFO | TWAP waiting 7.5 seconds for next interval...
2026-01-29 17:18:27 | INFO | TWAP Interval 2/4: Placing Market Order for 0.005
2026-01-29 17:18:27 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:18:27 | INFO | Market order placed successfully: 12022280315
2026-01-29 17:18:27 | INFO | TWAP Interval 2 placed. Total progress: 0.01/0.02
2026-01-29 17:18:27 | INFO | TWAP waiting 7.5 seconds for next interval...
2026-01-29 17:18:35 | INFO | TWAP Interval 3/4: Placing Market Order for 0.005
2026-01-29 17:18:35 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:18:35 | INFO | Market order placed successfully: 12022281141
2026-01-29 17:18:35 | INFO | TWAP Interval 3 placed. Total progress: 0.015/0.02
2026-01-29 17:18:35 | INFO | TWAP waiting 7.5 seconds for next interval...
2026-01-29 17:18:43 | INFO | TWAP Interval 4/4: Placing Market Order for 0.005
2026-01-29 17:18:43 | INFO | Placing MARKET order: BTCUSDT BUY 0.005
2026-01-29 17:18:43 | INFO | Market order placed successfully: 12022282429
2026-01-29 17:18:43 | INFO | TWAP Interval 4 placed. Total progress: 0.02/0.02
```

KEY ACHIEVEMENTS

✓ Fully functional CLI application
✓ Market and Limit orders working
✓ TWAP algorithmic execution (bonus)

✓ Real-time validation against exchange filters
✓ Comprehensive error handling
✓ Complete logging with timestamps
✓ Clean, modular code architecture
✓ Development time: ~30 minutes

CONCLUSION

Successfully implemented a production-ready trading bot with core functionality (market/limit orders) and advanced features (TWAP execution). The bot demonstrates efficient problem-solving, clean code organization, and thorough testing.

All code and documentation available at:
https://github.com/Vedant-D04/Binance-Trading-Bot.git